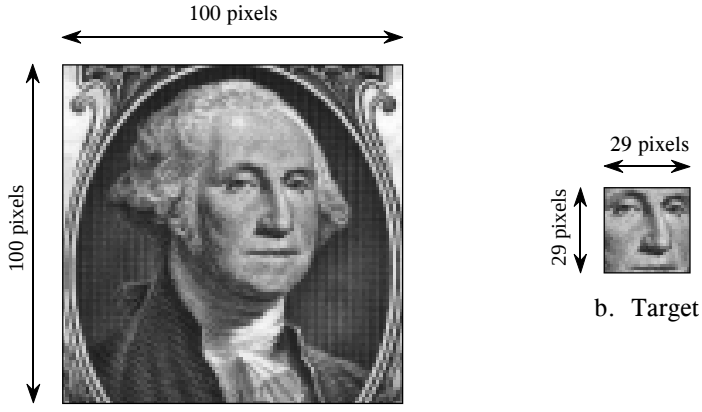

FIGURE 24-11
Target detection example. The problem is to search the 100×100 pixel image of George Washington, (a), for the target pattern, (b), the 29×29 pixel face. The optimal solution is *correlation*, which can be carried out by *convolution*.

before the correlation is performed. From the associative property of convolution, this is the same as applying the edge detection filter to the target signal *twice*, and leaving the original image alone. In actual practice, applying the edge detection 3×3 kernel only once is generally sufficient. This is how (b) is changed into (c) in Fig. 24-12. This makes (c) the PSF to be used in the convolution

Figure 24-13 illustrates the details of FFT convolution. In this example, we will convolve image (a) with image (b) to produce image (c). The fact that these images have been chosen and preprocessed to implement *correlation* is irrelevant; this is a flow diagram of *convolution.* The first step is to pad both signals being convolved with enough zeros to make them a power of two in size, and big enough to hold the final image. For instance, when images of 100×100 and 29×29 pixels are convolved, the resulting image will be 128×128 pixels. Therefore, enough zeros must be added to (a) and (b) to make them each 128×128 pixels in size. If this isn't done, circular

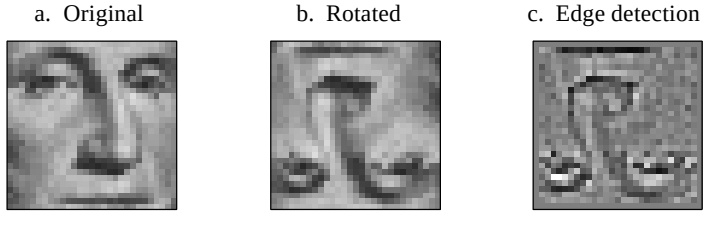

FIGURE 24-12
Development of a correlation filter kernel. The target signal is shown in (a). In (b) it is rotated by 180° to undo the rotation inherent in convolution, allowing correlation to be performed. Applying an edge detection filter results in (c), the filter kernel used for this example.